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NOTE

Soybean Biomass and Nitrogen Accumulation Rates and Radiation Use Efficiency in a Maximum Yield Environment

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ABSTRACT

Soybean [*Glycine max* (L.) Merr.] physiological characterization in a maximum yield environment may identify yield-optimization factors, lead to reassessment of fundamental crop model parameters, and provide guidance for management or breeding efforts. From 2011 to 2013, we characterized biomass and N accumulation rates, radiation use efficiency (RUE), and yield for four or five cultivars in a maximum-yield contest field. Grain yield among cultivars ranged from 5290 to 7953 kg ha^{-1}. The highest yields were observed in 2013, when biomass and N accumulation rates ranged from 45.6 to 64.3 g m^{-2} d^{-1} and 1.43 to 2.08 g N m^{-2} d^{-1}, respectively, and when RUE values ranged from 1.46 to 1.89 g MJ^{-1}. The observed crop growth characteristics in 2013 were near or above the maximum values previously reported in the literature. These empirical measurements provide collateral data documenting a soybean crop with grain yields approximately 6719 kg ha^{-1}.

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Abbreviations: DAE, days after emergence; FRI, fraction of canopy radiation interception; GDD, growing degree days; HI, harvest index; RUE, radiation use efficiency.

Data obtained from USDA-NASS (2013) indicate that average annual United States soybean grain yields have increased linearly from 739 kg ha^{-1} in 1924 to 2661 kg ha^{-1} in 2012 at a rate of 23 kg ha^{-1} yr^{-1}. Cassman et al. (2003) suggested that continued increases in average crop yield will depend on decreasing the difference between current and potential yields, which they defined as a "yield gap". Yield potential is defined as the "yield of a crop cultivar when grown with water and nutrients non-limiting and biotic stress effectively controlled" (van Ittersum et al., 2013, p. 5).

Soybean grain yield contests are currently conducted in 14 states in the United States (Alabama, Arkansas, Illinois, Kansas, Kentucky, Michigan, Missouri, North Carolina, Ohio, Pennsylvania, South Carolina, South Dakota, Virginia, and Wisconsin). The highest soybean yields reported in such contests to date were those submitted by Mr. Kip Cullers to the Missouri Soybean Association. He won their 2006, 2007, and 2010 Yield Contest with grain yields of 9400; 10,390; and 10,790 kg ha^{-1}, respectively (Cubbage, 2010). These are significant outlier yield contest values, given that, with just two exceptions, state soybean yield contest winners from 2000 to 2012 have not submitted yield entries exceeding 6719 kg ha^{-1} (100 bushels acre^{-1}). The exceptions to this are a 2007 Missouri contest yield of 7350 kg ha^{-1} (Steever, 2008) and a 2012 Michigan contest yield of 6751 kg

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ha^{-1} (Reinholt, 2012). These unusually high grain yields have created some controversy and due skepticism (e.g., Sinclair and Cassman, 2004). Some of this skepticism may be based on the potential uncertainty in what constitutes a theoretical maximum yield potential of soybean, but most skepticism is based on the lack of empirical collateral data that are context-specific and collected during the same growing season. Such data are important for providing evidentiary support of the exceptionally high yield claim.

Before 1966, it was commonly believed that the maximum soybean grain yield potential was near 4500 kg ha^{-1} (Cooper, 2003). Since that time, several yields > 6719 kg ha^{-1} have been reported in contests and scientific research (Cooper, 2003). Potential soybean grain yield estimates based on crop models range from 8290 kg ha^{-1} in Australia (Muchow and Sinclair, 1986), 5100 kg ha^{-1} for the 30-yr mean in Japan (Spaeth et al., 1987), and 5400 kg ha^{-1} in India (Bhatia et al., 2008). Moreover, if one assumes an RUE of 1.1 g MJ^{-1}, seasonal total interception of 1200 MJ m^{-2} of solar radiation, and a harvest index (HI) of 0.55, the inferred maximum yield is about 7300 kg ha^{-1} (Sinclair, 2004).

Specht et al. (1999) documented a 2.8-to-1 ratio between the rates of on-farm corn (*Zea mays* L.) and soybean grain yield improvement in 25 yr of Nebraska's irrigated soybean production systems. Egli (2008a) subsequently documented that the ratio of on-farm corn and soybean yield improvement in Iowa, Illinois and Indiana plateaued at about 2.9 to 3.0 after the early 1970s. Specht et al. (1999) used the 2.8 corn/soybean ratio to postulate that the maximum grain yield potential of soybean might be 8000 kg ha^{-1} based on the maximum observed corn grain yield from yield contests, near 22,500 kg ha^{-1} at that time, which was similar to the maximum corn yield potential suggested by de Wit (1967). Using this same logic, recently reported corn yields of 28,530 kg ha^{-1} (Wojcicki, 2013) would suggest that the maximum soybean yield potential should be adjusted upward to near 10,200 kg ha^{-1}. Regardless, Egli (2008b) concluded that it was not possible to know the true potential of soybean grain yield and that an apparently large amount of yield remained exploitable.

While not used in their yield-gap analyses, Fischer and Edmeades (2010) suggested that contest-winning crops are worthy of further study and may lead to identification of novel management practices, adjustments to the limits for parameters used in simulation models, or physiological processes to target for breeding efforts. The fundamental building blocks of soybean grain yield are carbohydrates, oil, and protein; thus, the maximum yield potential of soybean will ultimately be limited by C and N accumulation. Obtaining and sustaining high rates of photosynthetic activity requires high rates of N accumulation, and consequently, C and N accumulation are often highly correlated (Sinclair, 2004). Radiation use efficiency is an empirically estimable parameter that reflects the crop's ability to use solar energy for production of crop mass, and is typically measured as the ratio of crop biomass accumulated in a given period over energy input (intercepted solar radiation) during the same timeframe (Sinclair and Muchow, 1999). Using these concepts, a relatively simple model of soybean growth was developed and described by Sinclair (1986; Sinclair et al., 2003). In this model, the nominal maximum RUE values and daily N uptake for soybean are limited to 1.2 g MJ^{-1} and 0.6 g N m^{-2} d^{-1}, respectively.

Research was undertaken in Mr. Cullers' contest fields to independently generate replicated yield estimates and to characterize the biomass and N accumulation rates and RUE of the crop. The goals of this work were to provide physiological insight into soybean biomass and N accumulation rates, radiation use efficiency, and yield in a maximum yield environment, and to determine potential parameter limits in simulation models.

MATERIALS AND METHODS

Measurements were taken from Mr. Cullers' yield contest fields near Stark City, MO (36°51′ N, 94°11′ W) from 2011 to 2013. Five cultivars (Pioneer 94Y81, 94Y82, 94Y91, 94Y92, and 95Y10; DuPont Pioneer, Johnston, IA) in 2011, four cultivars (Pioneer 94Y23, 94Y80, 94Y81, and 94Y82) in 2012, and five cultivars (Pioneer 94B73, 48T53, 49T97, 50T40, and Asgrow 5332; Monsanto Co., St. Louis, MO) in 2013 were planted in blocks spanning the contest field. The plant rows aligned in an east–west direction, with the entirety of one cultivar being planted in a block, or strip, several planter-passes wide and consisting of approximately 3 ha. Each contest field was approximately 13 to 18 ha in total size. All cultivars had an indeterminate stem growth habit, and were glyphosate resistant with relative maturities of 4.2 to 5.3.

Two contest fields were utilized and the predominant soil type for both fields was a Newtonia silt loam (fine-silty, mixed, superactive, thermic, Typic Paleudolls). The two contest fields were in a corn–soybean rotation such that soybean was grown in Field A in 2011, Field B in 2012 and then back to Field A in 2013. A twin-row planter with Sync-Row units (Monosem Inc., Edwardsville, KS) was used to achieve a planting pattern in which the center of one twin-row pair (spaced 24-cm apart) was 76 cm from the center of an adjacent twin-row pair. The planting dates were 9 May 2011, 11 Apr. 2012, and 27 May 2013. The emergence (VE; Fehr and Caviness, 1977) dates were 3 to 13 d thereafter, on 16 May 2011, 24 Apr. 2012, and 30 May 2013. Four stand counts were taken soon after emergence from a 1-m length of a single twin-row pair from four locations within each cultivar block. Across cultivars, the mean plant density (± standard error) was 31.8 ± 0.6 plants m^{-2} in 2011, 30.6 ± 0.9 plants m^{-2} in 2012, and 25.0 ± 0.7 plants m^{-2} in 2013.

Each year since 2006 or before, poultry litter was applied and incorporated in the fall and again in the spring. Unfortunately, total manure applications, tillage operations, and fertilizer amounts were unavailable from the producer. Irrigation was provided through a center pivot system. Irrigation was applied frequently, often daily throughout the midseason, with approximately 1.3 cm of water per irrigation. Supplemental fertilizer was not included with irrigation during the timespan covering this report, in contrast to previous years. Irrigation applications, irrigation input totals, and soil test analyses were unavailable from the producer. Weeds were controlled with spring tillage coinciding with poultry litter incorporation and with one or more postemergence applications with glyphosate and other tank mixes as needed. Insects were controlled with multiple aerial applications of various insecticides, and fungal disease pressure was minimized by two or more prophylactic fungicide applications.

Successive aboveground biomass measurements were made in each cultivar block in each year. Samples were collected from four "plots", which were located approximately 1 m from the outside center pivot wheel track and evenly spaced across the cultivar block. In 2011, whole-plant biomass samples were collected from two sets of twin-row pairs at a length corresponding to a sampling area equivalent to 1 m^2 on 31 May, 15 d after emergence (DAE), when the fraction of canopy radiation interception (FRI) was approximately 0.10 and plants were at the V1 stage. A second set of 1-m^2 samples were collected at V6 on 13 June (28 DAE), when FRI was approximately 0.50, and a third set of 0.5-m^2 samples were collected at R1 on 27 June (42 DAE) when FRI was >0.92. In 2012, a first set of 0.5-m^2 samples were collected on 24 May at V6 (30 DAE) when FRI reached approximately 0.50. Additional sample sets were collected at R1 on 7 June (44 DAE), and at R2 on 19 June (56 DAE), when FRI was >0.93 in both instances. In 2013, the first 0.5-m^2 samples were collected on 27 June at V7 (28 DAE) when FRI reached approximately 0.65. The second sampling occurred on 11 July at R2 (42 DAE) and a third sampling set was taken on 24 July at R3 (55 DAE) when FRI for both dates was >0.95. Biomass samples were dried, weighed, ground to pass a 0.853-mm sieve, and analyzed for N concentration by the Dumas method with a Leco FP-428 Determinator (Leco Corp., St. Joseph, MO). The biomass accumulation rate (g m^{-2} d^{-1}) and N accumulation rate (g N m^{-2} d^{-1}) were calculated by plot as the respective difference between the total aboveground biomass and total N content for the last two biomass sampling dates, and divided by the days between those two measurements.

Daily total solar radiation and temperature were measured at the field perimeter with a silicon pyranometer and a shielded 12-bit temperature sensor coupled with a HOBO Micro Station (Onset Computer Corp., Pocasset, MA). Monthly rainfall data were obtained from NCDC-NOAA (2013a) from the Joplin, MO regional airport weather station, approximately 40 km from the field. Although rainfall at the field and irrigation applications and amounts were unavailable, it is assumed that the producer's frequent irrigation regime eliminated all water-deficit stress. Each year, canopy radiation interception was measured at least once every 7 d until FRI was >0.92, which occurred on 27 June 2011 (42 DAE), 7 June 2012 (44 DAE) and 4 July 2013 (35 DAE) as determined from digital-image analysis (Purcell, 2000). Linear regression of the weekly FRI measurements for each cultivar on accumulated growing degree days (GDD, using a base temperature of 10°C) generated a regression equation that was used to interpolate FRI values for each day within each of the 7-d intervals. Daily solar radiation measurements were then multiplied by the daily FRI to compute daily intercepted radiation values. Daily intercepted radiation values were successively summed to obtain a total cumulative radiation interception value for the period from VE to the day of the final biomass sample. Radiation use efficiency (g biomass MJ^{-1}) was determined for each plot by regressing the increase in aboveground biomass (g m^{-2}) against the cumulative amount of radiation intercepted by the crop (MJ m^{-2}) for all three biomass samples.

A few days after the last maturing cultivar reached harvest maturity (R8), 1-m^2 samples were collected from four bordered locations within each cultivar block near the biomass sample collection locations. These samples were used to determine final grain yield and HI, but no attempt was made to collect senesced leaves. Whole plant samples were dried, weighed, threshed, and the grain alone was dried overnight to remove any moisture gained, weighed, and adjusted to 130 g kg^{-1} moisture. Yield data and other variables measured during the season were analyzed using the MEANS procedure of SAS (SAS Institute, Cary, NC) to determine the cultivar means and standard errors for each variable in each year. Means were separated using a two-tailed *t* test and an $\alpha = 0.05$.

RESULTS

Soybean cultivar grain yields measured 5290 to 7137 kg ha^{-1} in 2011, 5521 to 6979 kg ha^{-1} in 2012, and 6158 to 7953 kg ha^{-1} in 2013 (Table 1). In 2011, the difference between the highest yielding cultivar and the third highest (6356 kg ha^{-1}) was not statistically significant ($\alpha = 0.05$). In 2012, the top two cultivars had nearly identical yields, whereas the bottom two also had nearly identical yields, but at a much lower yield level. The highest cultivar yield was observed in 2013 at 7953 kg ha^{-1}; however, greater yield variability was also observed in 2013. In 2013, only the lowest cultivar yield (6158 kg ha^{-1}) was significantly different from the greatest ($\alpha = 0.05$). Harvest index values ranged from 38.1 to 48.8%, with a mean of 43.9% over cultivars and years.

Across both the 2011 and 2012 growing seasons, the average temperatures for the months of June, July and August were 3.8, 5.8, and 2.8°C above the 30-yr mean, respectively (Table 2). However, 2013 was markedly cooler than the prior two seasons, and the mean maximum temperature never exceeded ±1.8°C from the 30-yr mean. Mean minimum temperatures were greater in 2011 than in 2012 and 2013 and were $\geq$ 3.1°C above the 30-yr mean throughout June and July. The mean daily solar radiation levels were near normal with the exception of May 2011, August 2012, and May and August 2013 being $\geq$ 2.7 MJ m^{-2} d^{-1} below the 30-yr mean. Although rainfall was below normal for periods in all 3 yr, frequent and timely irrigation eliminated drought as a confounding effect.

In 2011, biomass accumulation rates were measured between two sample dates when the FRI for the first date was approximately 0.50 and was >0.92 for the second date. Under these conditions, biomass accumulation rates among cultivars ranged from 13.6 to 18.2 g m^{-2} d^{-1} (Table 1). In 2012 and 2013, biomass accumulation rates were measured between two sample dates when FRI was >0.93 and >0.95, respectively. Biomass accumulation rates among cultivars ranged from 19.5 to 30.4 g m^{-2} d^{-1} (2012) and from 45.6

Table 1. Biomass and N accumulation rates, radiation use efficiency (RUE), grain yield, and harvest index (HI) means and standard errors (SE) by year and cultivar from Kip Cullers' contest field.

		Accumulation rate									
		Biomass		N		RUE		Yield		HI	
Year	Cultivar	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
		g m^{-2} d^{-1}		g m^{-2} d^{-1}		g MJ^{-1}		kg ha^{-1}		%	
2011†	94Y81	16.3 AB‡	± 0.9	0.58 A	± 0.05	0.90 A	± 0.03	7137 A	± 291	43.0 B	± 0.6
	94Y82	15.5 AB	± 1.4	0.52 A	± 0.07	0.96 A	± 0.09	7118 A	± 292	48.8 A	± 0.7
	94Y91	18.2 A	± 1.1	0.55 A	± 0.03	0.97 A	± 0.06	6117 B	± 200	44.0 B	± 0.8
	94Y92	13.6 B	± 0.4	0.47 A	± 0.03	0.93 A	± 0.03	5290 C	± 211	41.8 B	± 0.3
	95Y10	16.8 AB	± 2.0	0.58 A	± 0.05	1.03 A	± 0.11	6356 AB	± 298	43.3 B	± 0.8
2012§	94Y23	30.4 A	± 1.4	0.90 A	± 0.12	1.15 A	± 0.01	6979 A	± 193	48.2 A	± 1.2
	94Y80	25.6 AB	± 4.2	0.87 A	± 0.12	1.02 AB	± 0.10	6925 A	± 56	46.1 AB	± 0.4
	94Y81	19.5 B	± 1.1	0.83 A	± 0.04	0.84 B	± 0.02	5555 B	± 61	41.6 B	± 0.5
	94Y82	25.7 AB	± 3.2	0.88 A	± 0.10	1.01 AB	± 0.07	5521 B	± 119	45.5 AB	± 1.8
2013¶	94B73	60.6 AB	± 2.5	1.88 AB	± 0.17	1.89 A	± 0.06	7084 AB	± 489	41.4 B	± 1.1
	48T53	45.6 C	± 2.2	1.43 B	± 0.07	1.46 B	± 0.06	6158 B	± 175	38.1 B	± 2.6
	49T97	64.3 A	± 1.8	2.08 A	± 0.06	1.89 A	± 0.04	7953 AB	± 731	42.1 B	± 1.0
	50T40	60.6 AB	± 3.3	2.07 A	± 0.06	1.80 A	± 0.08	6883 AB	± 348	44.9 AB	± 0.3
	5332	56.2 B	± 1.0	1.51 B	± 0.17	1.83 A	± 0.04	7482 A	± 381	46.2 A	± 0.8

† RUE determined from three sample dates with corresponding canopy radiation interception of approximately 0.10, 0.50 and >0.92 and growth stages V1, V6, and R1. Biomass and N accumulation rates determined from last two sampling dates.

‡ Different letters within a column and year denote that means differed (α = 0.05) as determined by a two-tailed *t* test.

§ RUE determined from three sample dates with corresponding canopy radiation interception of approximately 0.50, >0.93, and >0.93 and growth stages V6, R1, and R2. Biomass and N accumulation rates determined from last two sampling dates.

¶ RUE determined from three sample dates with corresponding canopy radiation interception of approximately 0.65, >0.95, and >0.95 and growth stages V7, R2, and R3. Biomass and N accumulation rates determined from last two sampling dates.

Table 2. Mean monthly high temperature (Tmax), low temperature (Tmin), solar radiation (Rs) from Kip Cullers' farm and monthly total rainfall (NCDC-NOAA, 2013a) in 2011, 2012, and 2013. Departures from the 30-yr mean (1981–2010; NCDC-NOAA, 2013b) are in parentheses.

Year	Month	Tmax	Tmin	Rs†	Rainfall‡
		°C	°C	MJ m^{-2} d^{-1}	mm
2011	Apr.	21.8 (+1.6)	9.2 (+2.5)	19.6 (–1.5)	162 (+44)
	May	23.3 (–1.1)	12.7 (+0.5)	18.6 (–3.8)	26 (–126)
	June	32.3 (+3.7)	20.6 (+3.7)	24.4 (+1.5)	27 (–114)
	July	36.5 (+5.3)	22.5 (+3.1)	23.7 (+0.8)	38 (–55)
	Aug.	34.6 (+3.0)	20.7 (+2.1)	20.0 (–1.6)	126 (+42)
	Sept.	26.0 (–1.0)	11.6 (–1.9)	16.5 (–2.0)	124 (–6)
2012	Apr.	22.8 (+2.6)	10.9 (+4.2)	19.0 (–2.1)	159 (+41)
	May	28.8 (+4.4)	14.3 (+2.1)	21.7 (–0.7)	105 (–47)
	June	32.4 (+3.8)	17.0 (+0.1)	23.2 (+0.3)	32 (–109)
	July	37.5 (+6.3)	20.5 (+1.1)	22.5 (0.0)	0 (–93)
	Aug.	34.2 (+2.6)	17.4 (–1.2)	18.8 (–2.8)	80 (–4)
	Sept.	27.6 (+0.6)	15.4 (+1.9)	16.2 (–2.3)	183 (+53)
2013	Apr.	18.5 (–1.7)	6.7 (0.0)	19.6 (–1.5)	169 (+51)
	May	23.4 (–1.0)	12.2 (0.0)	19.7 (–2.7)	200 (+48)
	June	29.3 (+0.7)	18.4 (+1.5)	22.2 (–0.7)	137 (–4)
	July	30.8 (–0.4)	18.3 (–1.1)	20.5 (–2.0)	84 (–9)
	Aug.	30.4 (–1.2)	18.2 (–0.4)	18.5 (–3.1)	117 (+33)
	Sept.	28.8 (+1.8)	14.9 (+1.4)	17.2 (–1.3)	35 (–95)

† Solar radiation 30-yr means calculated with 30-yr mean high and low temperatures using a modified Hargreaves and Samani (1982) equation, described by Ball et al. (2004).

‡ Irrigation was applied frequently in an attempt to eliminate all water deficit stress; however total inputs were unavailable from the producer.

to 64.3 g m^{-2} d^{-1} (2013). With the same sampling dates, nitrogen accumulation rates among cultivars ranged from 0.47 to 0.58 g N m^{-2} d^{-1} (2011), 0.83 to 0.90 g N m^{-2} d^{-1} (2012) and 1.43 to 2.08 g N m^{-2} d^{-1} (2013).

Linear regression of the weekly FRI values versus the coincident accumulated GDDs from emergence provided a good fit with r^2 values $\geq$0.90 for all cultivars in all years (data not shown). Radiation use efficiency values were similar between 2011 and 2012 and ranged among cultivars from 0.84 to 1.15 g MJ^{-1}, with an average value of 1.01 g MJ^{-1}. In 2013, RUE values among cultivars ranged from 1.46 to 1.89 g MJ^{-1}, with an average over cultivars of 1.77 g MJ^{-1}.

DISCUSSION

Mr. Cullers won the 2011 Missouri Soybean Association yield contest with a yield of 7317 kg ha^{-1} for the Pioneer cultivar 94Y80 (Missouri Soybean Association, 2011), which was managed under a separate irrigation system (drip tape) in the same field but not under the center pivot or within our sampling area. This reported yield was similar to our 2011 yield estimates for the two highest yielding cultivars. Mr. Cullers did not enter the 2012 yield contest due to "low and inconsistent yields" over the required 1.62-ha harvest area. However, our 2012 1-m^2 yield estimates indicate that the two highest yielding cultivars were greater than the 6230 kg ha^{-1} yield that did win (Missouri Soybean Association, 2012). In 2013, Mr. Cullers' highest cultivar yields were approximately 7730 kg ha^{-1} (Kip Cullers, personal communication, 2013), which were also similar to the best two cultivars from our yield estimates. The foregoing yields are substantively less than the yields of 9400; 10,390; and 10,790 kg ha^{-1} that Mr. Cullers submitted as 2006, 2007, and 2010 entries to the MO soybean yield contests.

The reasons for the lower contest yields reported here relative to previous years are not known. Weather conditions in 2011 and 2012 were abnormally warm and dry compared with the 30-yr mean. The mean maximum temperature in July for both years was 37°C, averaging 5.8°C greater than the 30-yr mean (Table 2). For comparison, monthly average maximum temperatures near Stark City, MO in 2006, 2007, and 2010, were near normal during the growing season and were never >3.1°C of the 30-yr mean and with July maximum temperatures ranging no more than ±1.4°C from the 30-yr mean (data not shown). Gibson and Mullen (1996) demonstrated that increasing day temperatures from 30 to 35°C throughout reproductive growth (R1 to R8) decreased seed growth and photosynthetic rates and ultimately reduced seed yield plant^{-1} by 27%. Additionally, raising the temperature 2 to 3°C above ambient decreased soybean grain yield from 16 to 40% (Tacarindua et al., 2013). Conversely, other studies have demonstrated photosynthetic rates remained stable across a range of temperatures from 26 to 36°C (Campbell et al., 1990; Jones et al., 1985) and that leaf area, aboveground biomass production, and yield were similar between 31 and 24°C and 36 and 29°C temperature treatments despite a reduction in the HI with increasing temperatures (Baker et al., 1989).

In 2013, mean maximum temperatures were near normal and were never >1.8°C from the 30-yr mean (Table 2). However, due to cool April temperatures and frequent April and May rainfall, planting in 2013 did not occur until 27 May. This may have reduced the yield potential in 2013. De Bruin and Pedersen (2008) found in Iowa that planting in late April compared with early June decreased yield from 12 to 41%, with a mean yield reduction of 25%. The authors also suggested that yield decreases from delayed planting were greater in locations with high-yield potential than locations with lower-yield potential. Regardless of why yields of greater magnitude were not observed, the data presented here provide empirical measurements to describe the crop growth characteristics that resulted in yields of 5290 to 7953 kg ha^{-1} during three growing seasons.

The 2011 biomass and N accumulation rates were not representative of the maximum rates attainable for the crop because the first sampling date occurred before canopy closure. Biomass accumulation rates in 2012 were near or above the average of the maximum values that we could find reported for soybean in the literature (Supplemental Table S1). In Supplemental Table S1, we have listed the highest values of soybean biomass and N accumulation rates and RUE from the literature representing 31, 12, and 44 site years of data for these variables, respectively. These maximum rates of biomass accumulation ranged from 8.9 to 55 g m^{-2} d^{-1} and averaged 19.8 g m^{-2} d^{-1}. The 2013 biomass accumulation rates were similar to the maximum rate of 55 g m^{-2} d^{-1} previously reported in the literature (Isoda et al., 2010).

As expected, the high biomass accumulation rates observed in our research resulted in high N accumulation rates as well. The N accumulation rates in 2012 were 4 to 13% greater than the highest value reported for soybean in the literature, which ranged from 0.31 to 0.80 g N m^{-2} d^{-1}, with a mean of 0.45 g N m^{-2} d^{-1} (Supplemental Table S1). In 2013, the N accumulation rate averaged over cultivars was 1.79 g N m^{-2} d^{-1} and was more than double that of the previously documented maximum N accumulation rate for soybean. Using a non-nodulating reference cultivar, the fraction of N derived from atmospheric N_2 was determined to range from 0 to 17%, depending on cultivar and sampling date, and averaged 7% across cultivars and sampling dates in 2011 and 2013 (data not shown).

The highest soybean biomass accumulation rates (35 to 55 g m^{-2} d^{-1}) were reported by Isoda et al. (2010) and resulted in extremely high yields, up to 9200 kg ha^{-1}. Management by Isoda et al. (2010) also included perennial poultry litter applications (15 Mg ha^{-1} yr^{-1}) and high irrigation inputs similar to what we observed at Mr. Cullers' farm. Egli and Zhen-wen (1991) reported a soybean biomass accumulation rate of 27.0 g m^{-2} d^{-1} for measurements made in China; however, the biomass accumulation rate was accompanied by a relatively small amount of photosynthate being partitioned to seed growth, presumably resulting in a low HI. The reason for the low photosynthate partitioning to seed was not determined, but may explain why soybean grain yield in that study (≤3540 kg ha^{-1}) lagged behind the yields observed by Isoda et al. (2010) and in our research. Pal and Saxena (1976) also reported a large N accumulation rate (0.80 g N m^{-2} d^{-1}), but the highest yield reported was 3220 kg ha^{-1} and no indications were provided to explain why higher yields were not observed.

The high rate of N accumulation and high RUE that we found in 2013 did not appear to increase yields compared with 2011 and 2012. We speculate that the biomass and N accumulation rates and RUE values observed in 2013 during vegetative and early reproductive development were probably not maintained throughout reproductive development based on yield and HI data. If an RUE of this magnitude could be sustained throughout seed fill, grain yields similar to those reported by Isoda et al. (2010; 9200 kg ha^{-1}) and by Mr. Cullers in 2010 (10,790 kg ha^{-1}) may be possible.

The maximum soybean RUE values reported in the literature for a single treatment or cultivar in each site-year that we reviewed ranged from 0.5 to 1.36 g MJ^{-1}, with a mean of 0.85 g MJ^{-1} (Supplemental Table S1). The RUE

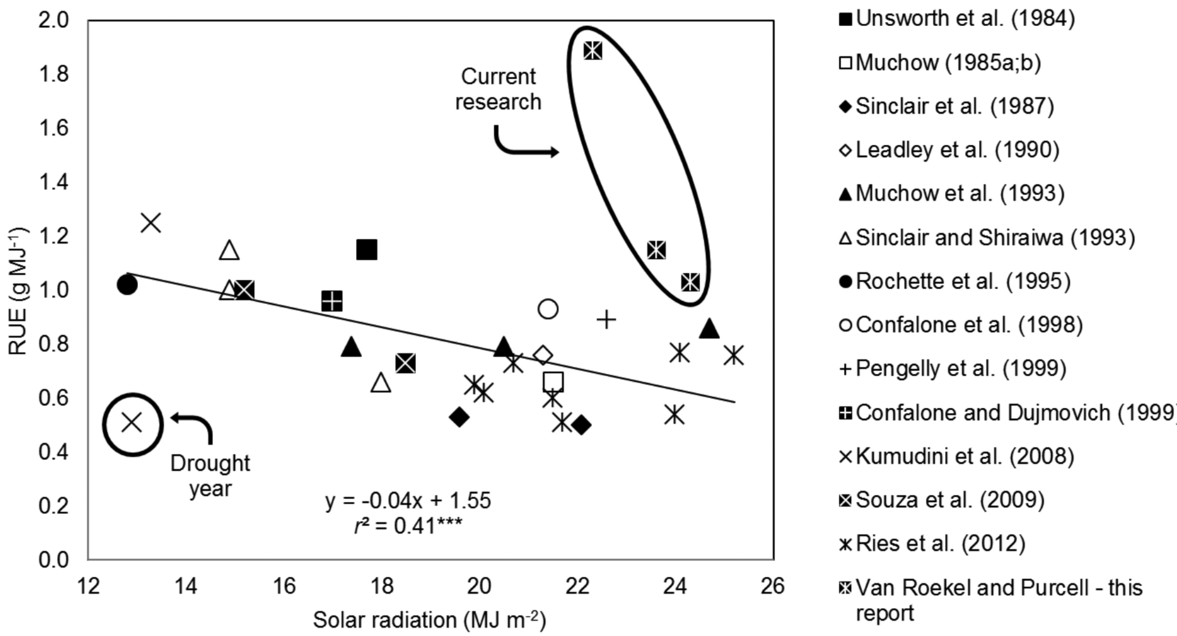

Figure 1. Maximum reported radiation use efficiency (RUE) within each site-year versus mean daily solar radiation during the time of sampling from available literature sources as well as measurements from the current research in 2011, 2012, and 2013. The circled values, from the current research, were not included in the regression nor were the 2005 to 2006 data from Kumudini et al. (2008) due to drought conditions.

values recorded in this research in 2011 and 2012 were generally above the mean value of 0.85 g MJ^{-1} (Table 1). In 2013, RUE values of all cultivars (1.46 to 1.89 g MJ^{-1}) were greater than the previously reported maximum soybean RUE. For comparison, Nakaseko and Gotoh (1983) reported a soybean RUE of 1.36 g MJ^{-1}, and Kumudini et al. (2008) reported a RUE of 1.26 g MJ^{-1} from research in 2006–07.

Sinclair et al. (1992) postulated that RUE values would increase with decreasing mean daily radiation because leaf photosynthetic rates become less efficient as radiation levels approach the light saturation level (Monteith, 1977; Sinclair and Horie, 1989). For example, Manrique et al. (1991) documented the RUE for potato (*Solanum tuberosum* L.) decreased 0.15 g MJ^{-1} for every MJ m^{-2} d^{-1} increase in total solar radiation from 12 to 26 MJ m^{-2} d^{-1}. Indeed, a negative trend was found on reviewing the values for RUE and solar radiation in the soybean literature (Supplemental Table S1; Fig. 1). Each value cited in Supplemental Table S1 was "cherry picked" as the highest value for a single treatment or cultivar within each site-year of the research; thus, only the highest RUE value from one cultivar in each year of this research was included in Fig. 1. Linear regression indicated that RUE values decreased by 0.04 g MJ^{-1} for every MJ m^{-2} d^{-1} increase in the mean daily solar radiation. The mean RUE value of the data in Fig. 1 was 0.80 g MJ^{-1} at a mean solar radiation level of 19.6 MJ m^{-2} d^{-1}. Despite having relatively high average solar radiation levels during the RUE measurement period (24.3, 23.6, and 22.3 MJ m^{-2} d^{-1} in 2011, 2012, and 2013, respectively), RUE values in 2011 and 2012 were among the highest values reported in the literature across all solar radiation levels. Averaged across cultivars, the RUE we found in 2013 was 30% greater than the highest RUE previously reported. On the basis of the the regression equation in Fig. 1 and the observed solar radiation, the predicted RUE values were 0.58 (2011), 0.61 (2012), and 0.66 (2013) g MJ^{-1}, while the actual RUE values were 78, 90, and 186% greater than these predicted values.

The physiological characteristics presented here represent unique empirical measurements of a crop grown within a maximum yield environment. Some of these measurements represent new upper limits compared with previous reports in the literature. In the Sinclair soybean model (Sinclair, 1986; Sinclair et al., 2003), N accumulation during vegetative growth is based on the N demand of the expanding leaf area and stem mass, and the maximum N accumulation rate is nominally limited at 0.6 g N m^{-2} d^{-1}. While this value of N accumulation is above the mean of maximum values previously reported for soybean (Supplemental Table S1), it is about 30% of the maximum N accumulation rate found in our research (2.08 g N m^{-2} d^{-1}). Additionally, the nominal maximal RUE value (1.2 g MJ^{-1}) in the model was 63% of the maximal RUE value found in our research (1.89 g MJ^{-1}). High RUE values are predicted to only be maintained as long as leaf N levels remain high, which depends on high rates of N accumulation. In the Sinclair soybean model, large daily increases in modeled biomass production may result in a dilution of leaf N resulting in decreased RUE if the maximum N accumulation rate is also underestimated. This is one example of how modeling efforts may need to be re-examined or adjusted when predicting soybean yield in maximum yield environments with abundant soil N.

CONCLUSIONS

The physiological characteristics observed in this research provide empirical data describing a soybean crop with approximately 6719 kg ha^{-1} (i.e., 100 bushel acre^{-1}) yield potential. The biomass and N accumulation rates and RUE values observed in 2013 were greater than previously reported values for soybean. We can infer from these measurements how sufficient biomass was produced to support these yields at the observed HI values. While the data collected in this research cannot be used to make definitive inferences about yields greater than the observed yield range, it does offer physiological insight into the N accumulation rate and RUE necessary for the production and modeling of soybean yields up to 7953 kg ha^{-1}.

Supplemental Information Available

Supplemental information is included with this article.

Summary of the maximum reported values in the literature from a single treatment or cultivar within each site-yr for the rate of biomass and N accumulation of soybean, radiation use efficiency (RUE), and mean daily solar radiation (Rs) by source, location, and year of research.

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